

Group 10 Homework 2

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1 Training & Deployment of a “Up/Down” Keyword Spotter

1.1 Preprocessing hyperparameters search space

In order to find the hyperparameters combination compliant with the constraints of **Accuracy**, **TFLite Size** and **Total median latency on the RPI**, a grid-search approach was taken over the following search spaces:

- **frame length (in ms)**: the space used for the frame length is $D=\{16, 32\}$. Note that they're powers of 2 and chosen such that they are near the average length of an english phoneme ($[10, 50]$)
- **frame step (in ms)**: the frame step depends on the value of the frame length since it is chosen such that the overlap is in $\{0\%, 25\%, 50\%, 75\%\}$ with respect to the frame length value.
- **# of Mel bins (int)**: this parameter indicates the number of frequency bands in which the spectrum is divided. A higher value guarantees a higher level of detail on the frequencies but can lead also to an overfitting risk, raising the dimensionality of the data. In our case, the search space was in $\{30, 40\}$ in order to find balance between good generalization and overfitting.
- **lower frequency (in Hz)**: it is the minimum frequency of the range window which will be considered for the signal transformation. The search space was in $\{20, 40, 60\}$.
- **upper frequency (in Hz)**: it is the maximum frequency of the range window which will be considered for the signal transformation. The search space was in $\{2000, 4000, 6000\}$.
- **# of coefficients (int)**: this parameter was inserted in the grid search in order to include also the **MFCC** technique; when its value is equal to 0, the model adopts a pure **Mel Spectrogram** approach, while if it is positive the model adopts **MFCC**. In particular, this number determines how many coefficients (from the Discrete Cosin Transform applied on the logarithmic energies on the Mel bands) will be kept, so the higher its value the more spectrum details are kept (also in this case there is an overfitting risk). The search space was in $\{0, 30, 40\}$.

1.2 Final solution

The grid search operation on the hyperparameters, which was adopted mainly to satisfy the accuracy constraint, gave as result not only one but a few combinations, so only one of them was selected for the latency testing operation on the RPI. In order to reduce the TFLite model size, the adopted methods are described in the section 1.3 . In the tables [1],[2] and [3] the hyperparameters and the results can be observed.

Frame length (ms)	Frame step (ms)	# of Mel bins (int)	Lower frequency (Hz)	Upper frequency (Hz)	# of coefficients (int)
16	16	30	20	6000	30

Table 1: Preprocessing hyperparameters of the final solution

Batch size (int)	learning rate (float)	end learning rate (float)	# of epochs (int)	Width multi- plier (float)
20	10^{-2}	10^{-4}	20	0.5

Table 2: Training hyperparameters of the final solution

Accuracy (%)	TFLite size (KB)	total median latency (ms)
99.500	48.2	25.4 +/- 0.3

Table 3: Results of the final solution

1.3 Model architecture and optimizations

The model was built as a sequential one with three Conv2D layers and two DepthwiseConv2D layers, as shown in the image (1). This alternation of layers reduces complexity with a limited impact on accuracy. In order to accomplish the size constraints, structured pruning via weight multipliers was adopted, halving the number of filters for each convolutional layer.

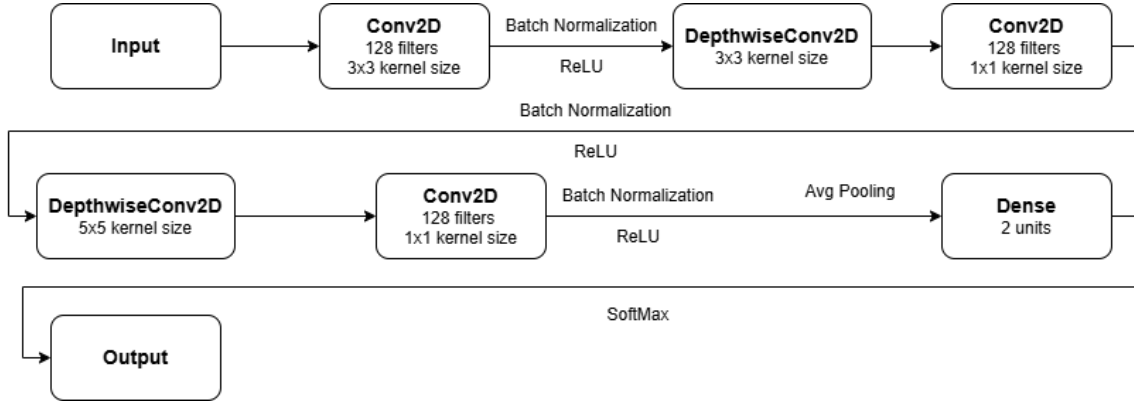


Figure 1: Model architecture

1.4 Comments

The developed model respects all the requested constraints thanks to weights pruning (on the model size side) and depth-wise layers (on the latency side). The kernel size of the last depth-wise layer was increased to capture more general features and this led to a higher accuracy. The configuration chosen with respect to accuracy and model size also satisfied the latency constraint without any further modification.